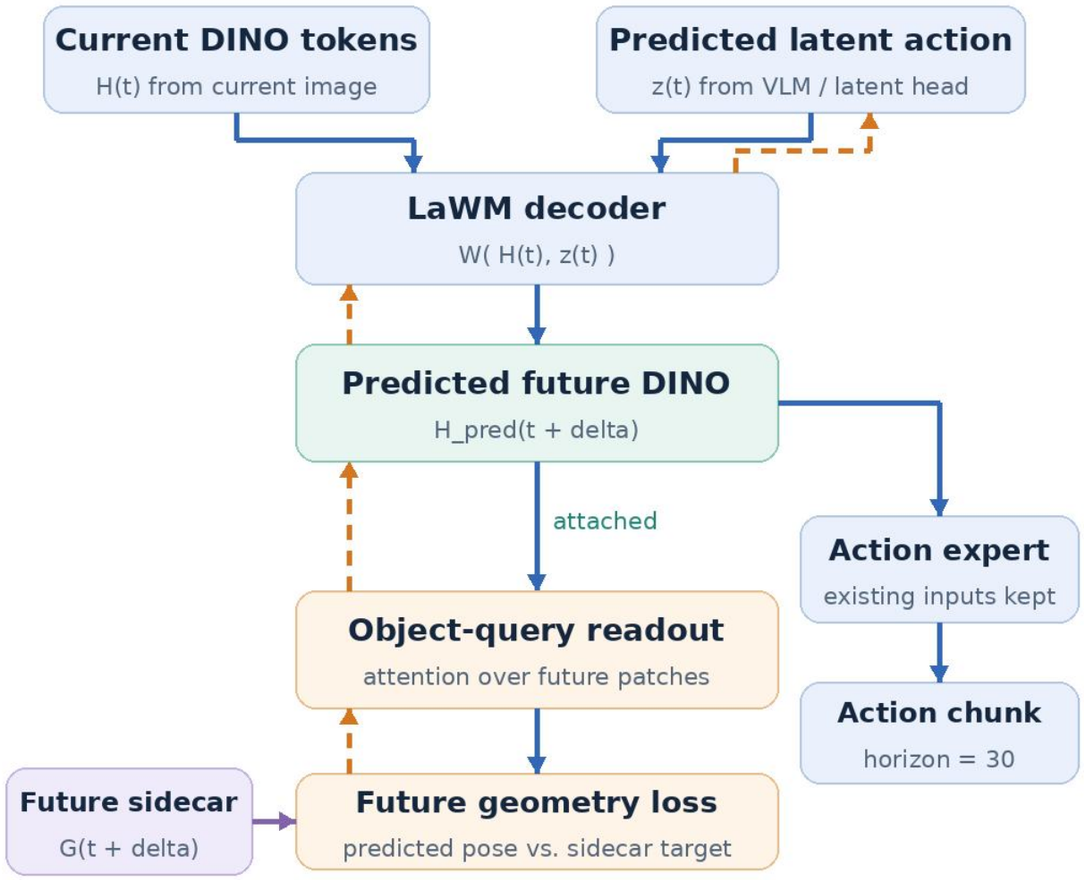
Future object geometry as supervision for LaWM

Proposed controlled experiment | Same forward values at fixed weights; different auxiliary gradients

visual_wm_aux

Geometry loss shapes the future representation



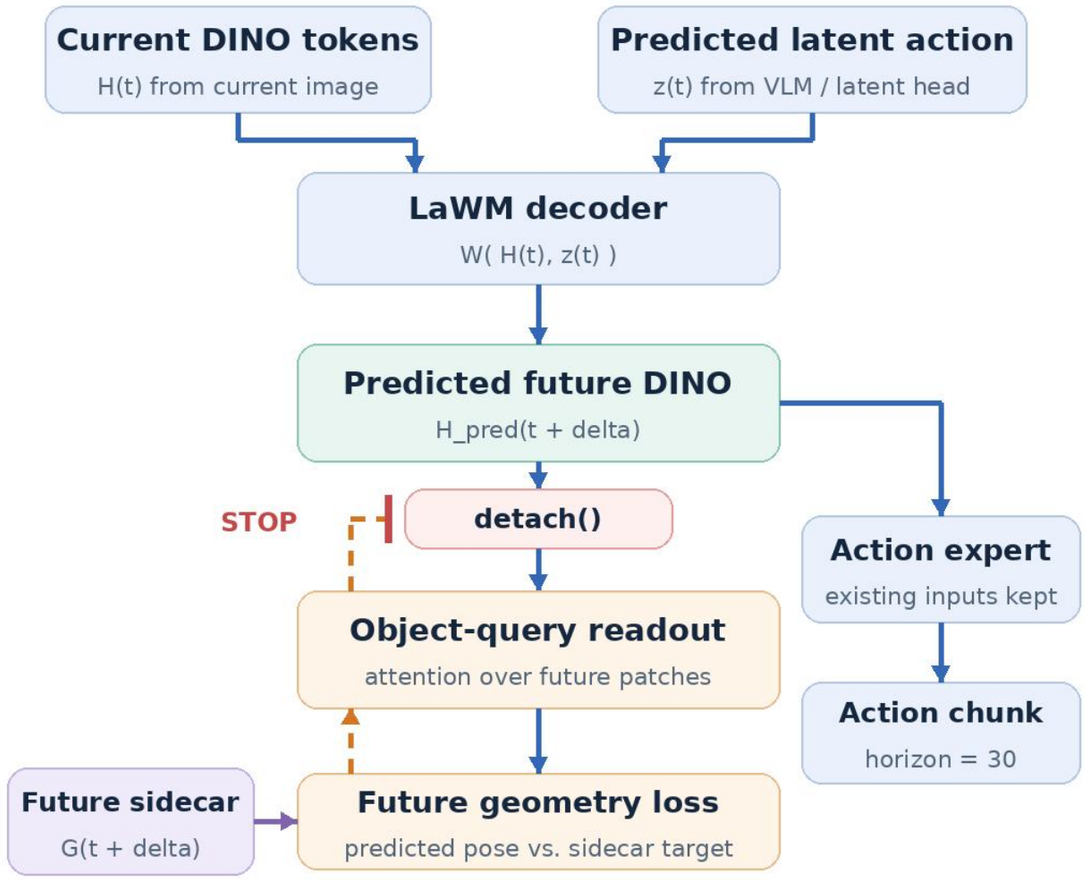
L_geo updates readout + decoder + latent branch.

The action expert consumes the shaped future features.

Training-only branch: readout + geometry loss + sidecar.
At inference: keep the original LaWM-to-action path.

visual_wm_aux_detach

Readout-only auxiliary control



L_geo updates the readout only.

No L_geo gradient enters LaWM or the latent head.

Training-only branch: readout + geometry loss + sidecar.
At inference: keep the original LaWM-to-action path.

Forward computation | Geometry-loss gradient only | Training label supervision

Both: original Stage 2 losses remain active. Detach does NOT freeze the policy or detach the action branch.

Future labels match the sampled future-image timestamp. Other existing conditioning and loss paths are omitted.